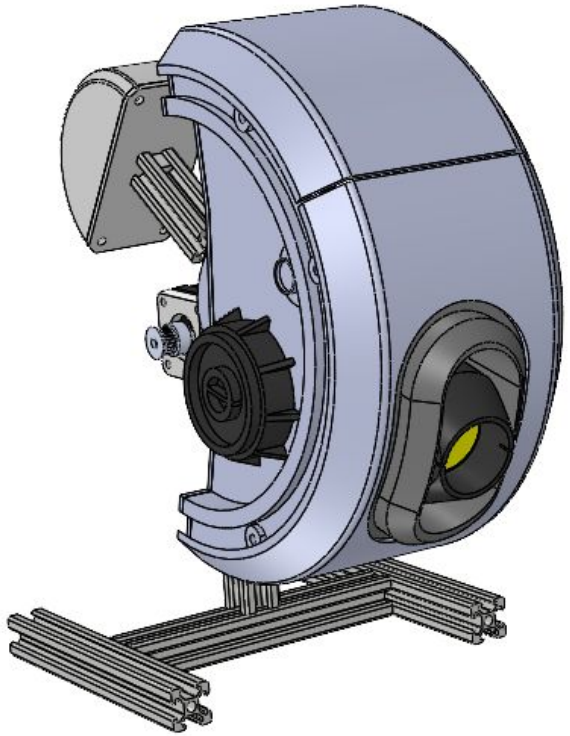
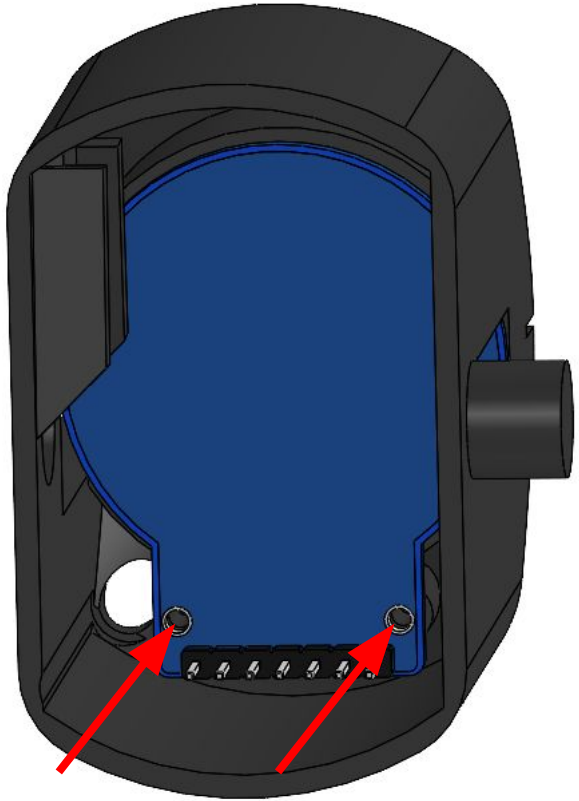
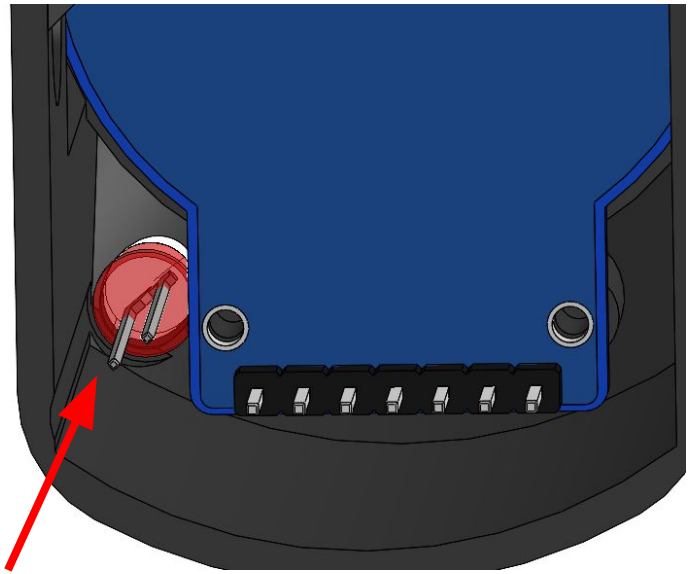
Crafitys Glados

Assembly instruction

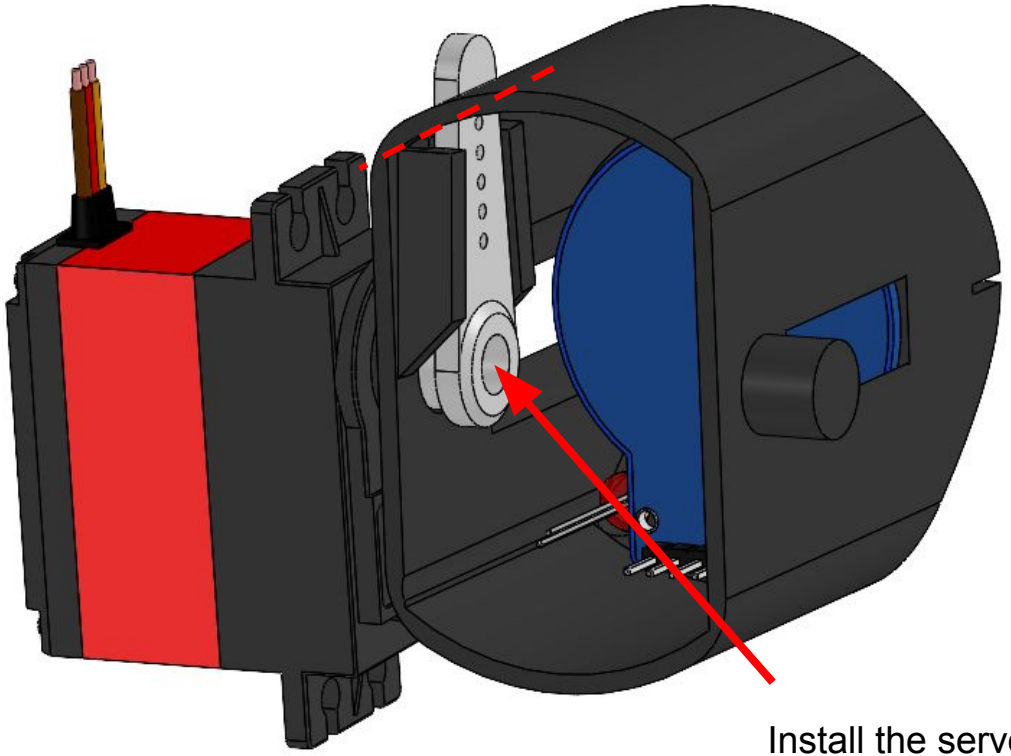




Secure the round display using two Ø2 × 6mm plastic screws

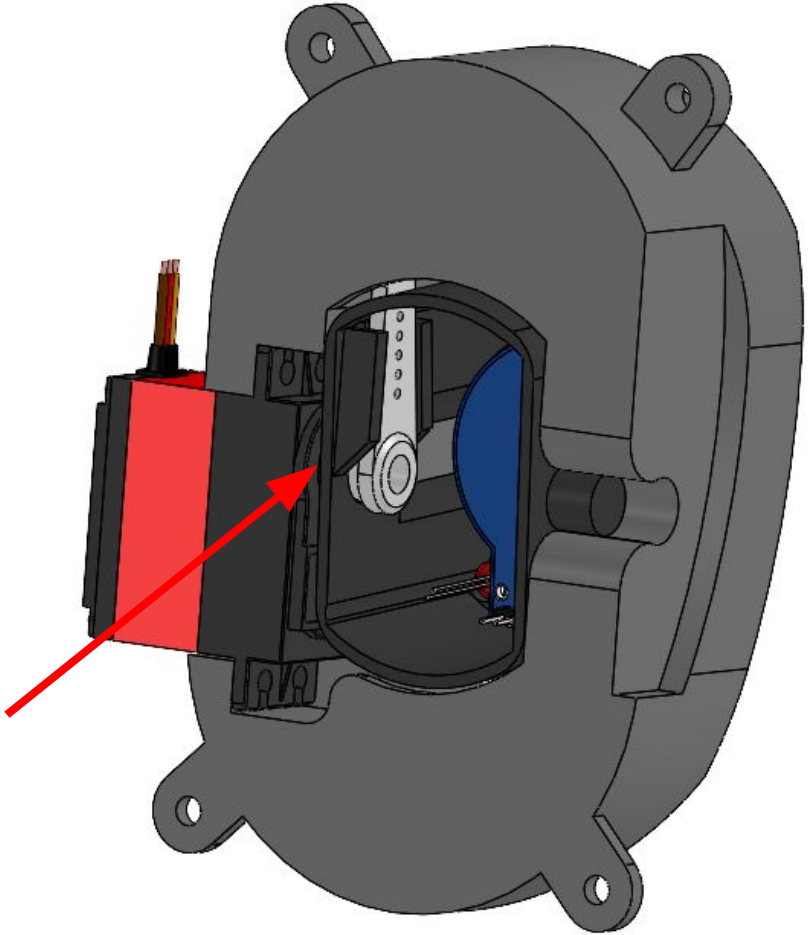


Insert the red LED with two wires pre-soldered, then secure it in place using a hot glue gun

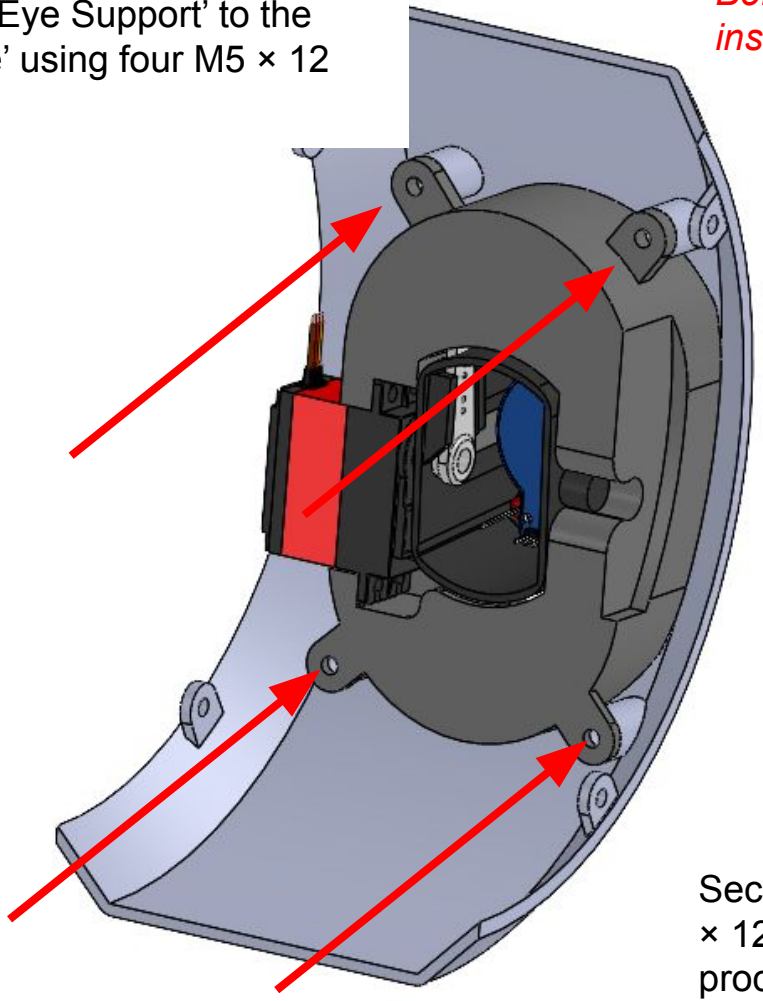


Install the servo motor and its attachment. Remember to trim the part that interferes with the eye

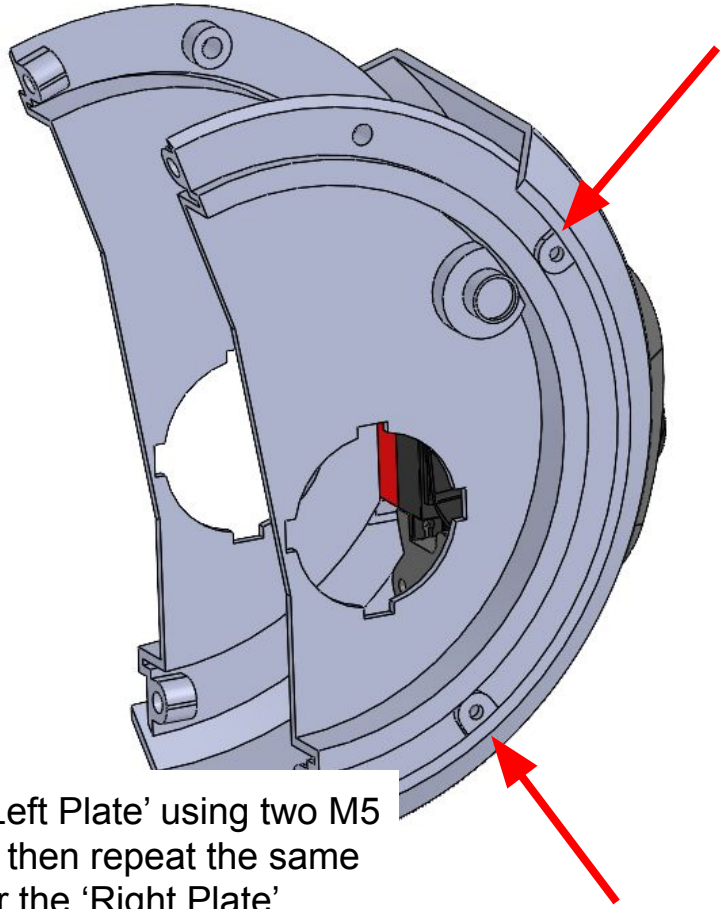
Install the assembly into the ‘Eye Support’ part and secure the motor in place



Secure the ‘Eye Support’ to the ‘Lower Plate’ using four M5 × 12 screws

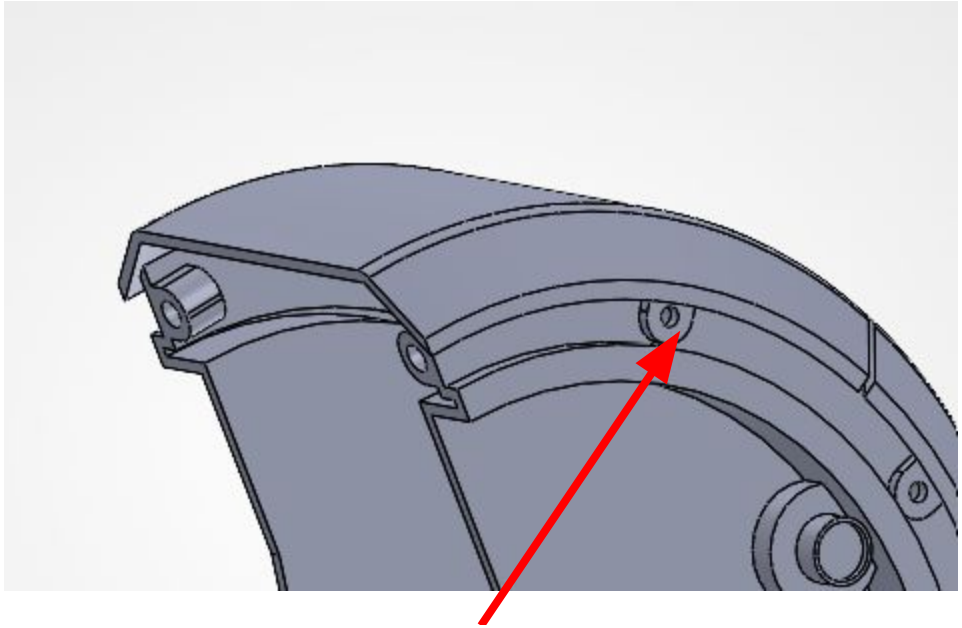


Before installing the screws, insert the M5 threaded inserts into their slots using a soldering iron

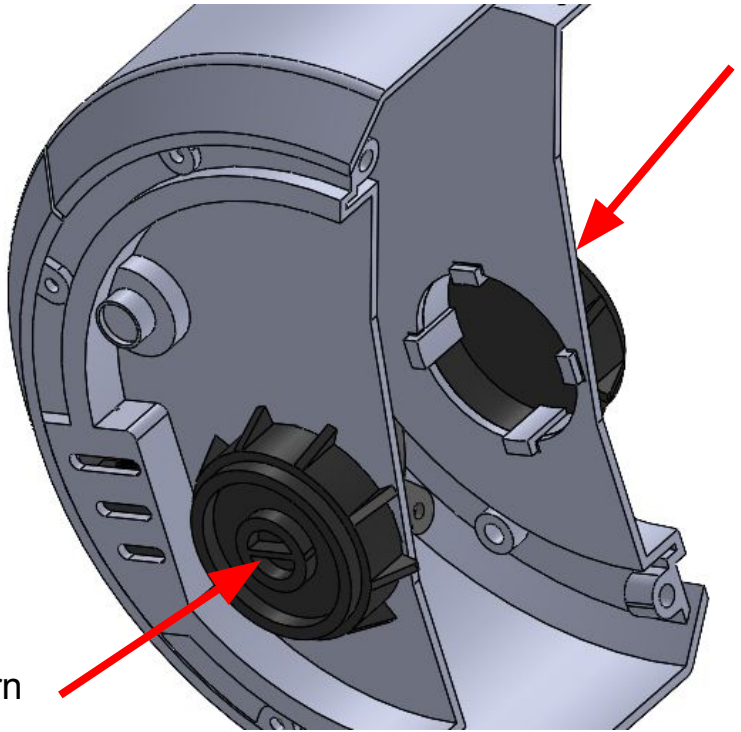


Secure the ‘Left Plate’ using two M5 × 12 screws, then repeat the same procedure for the ‘Right Plate’

Before installing the screws, insert the M5 threaded inserts into their slots using a soldering iron

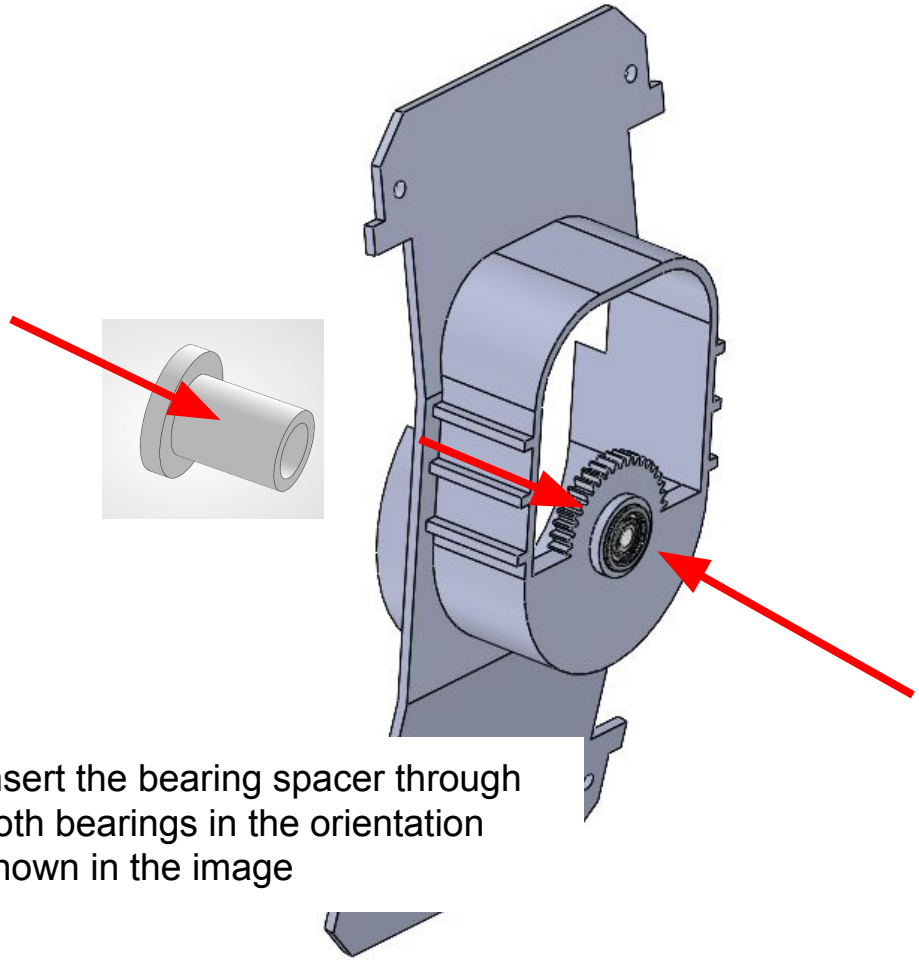


Secure the ‘upper Plate’ using two M5 × 12 screws

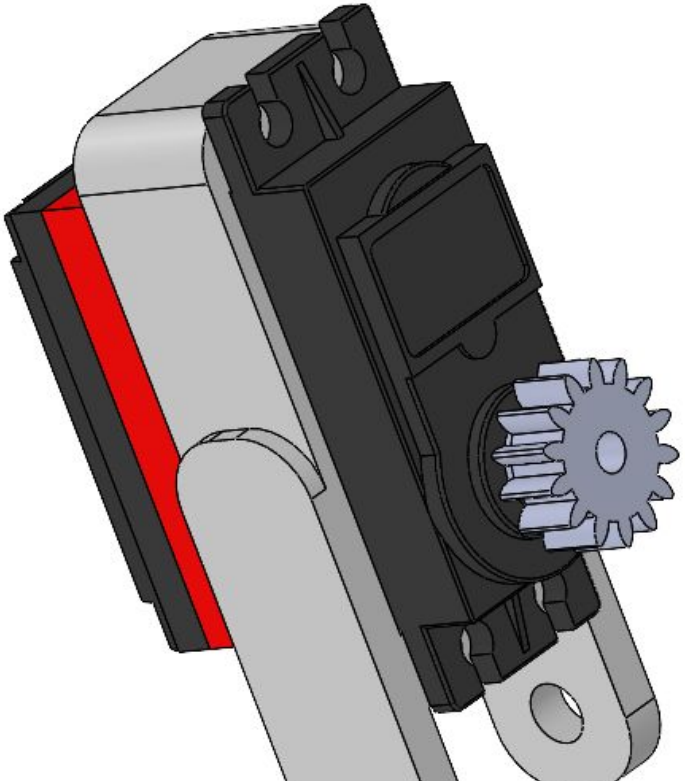


Attach both ears and lock them in place by turning them a quarter turn

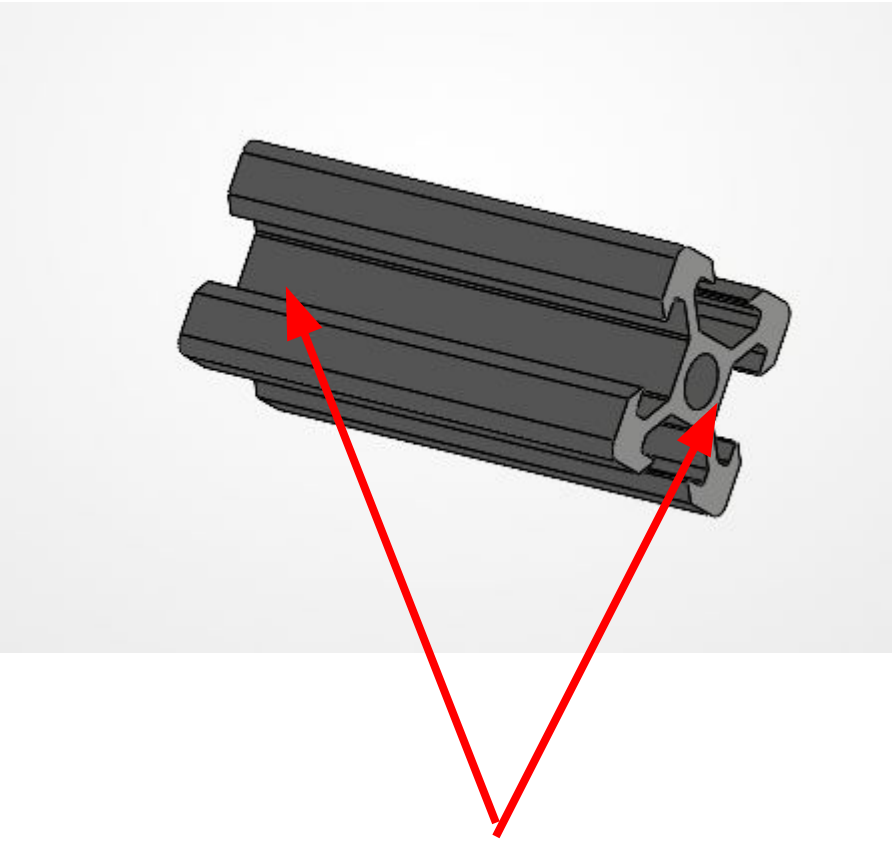
Install two F688ZZ bearings



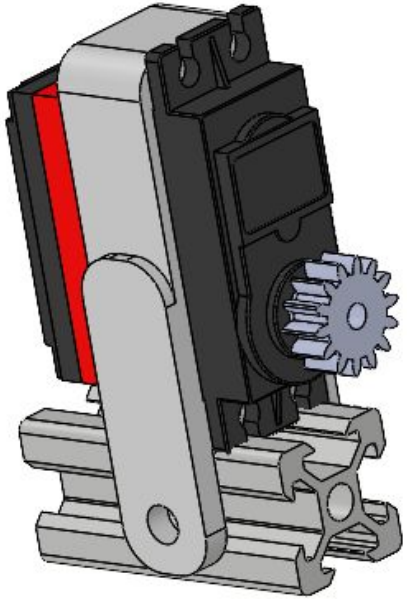
Insert the bearing spacer through both bearings in the orientation shown in the image



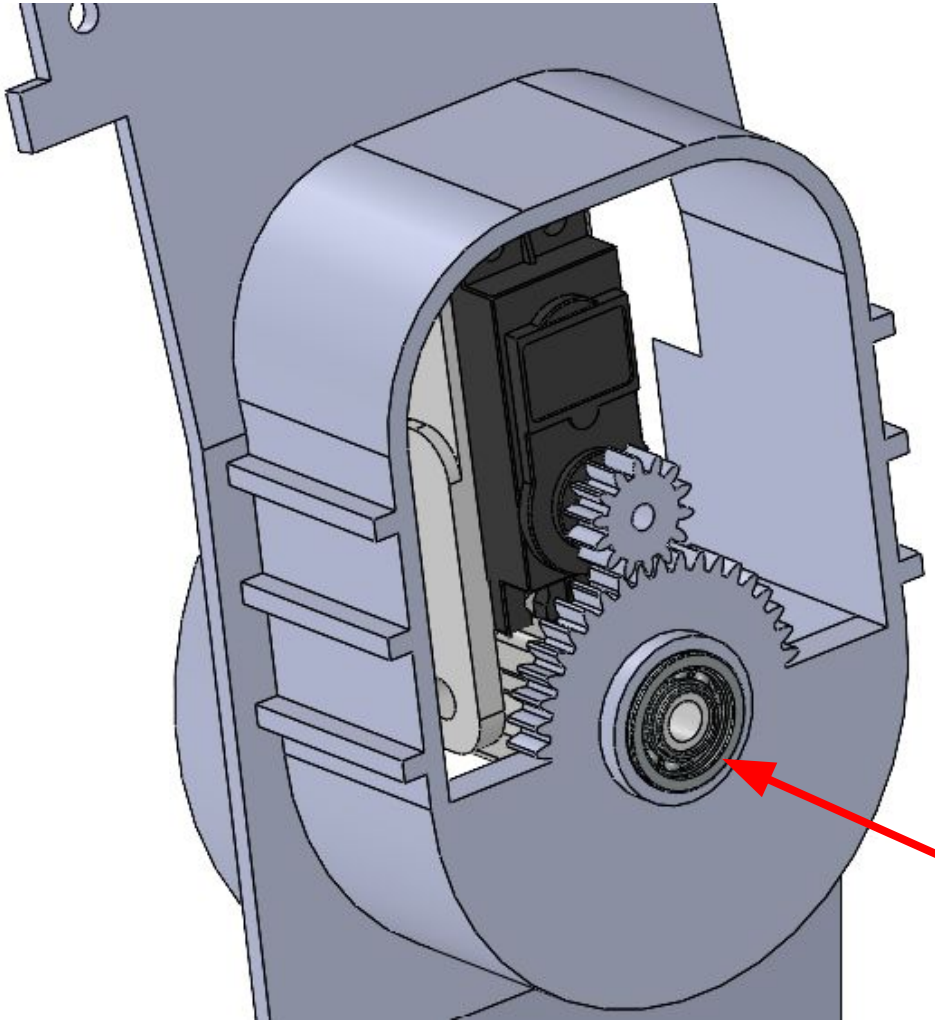
Secure the head tilt servo motor to its bracket using the four screws, then attach the gear to the servo shaft using the appropriate screw supplied with the motor

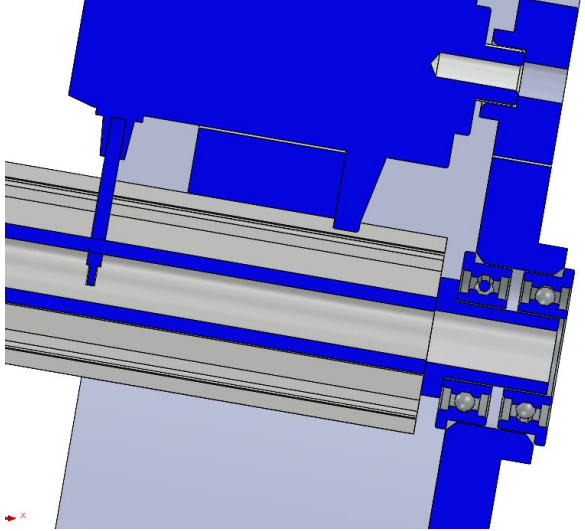


Cut a 47 mm length of 2020 aluminum extrusion and tap both ends with an M5 thread



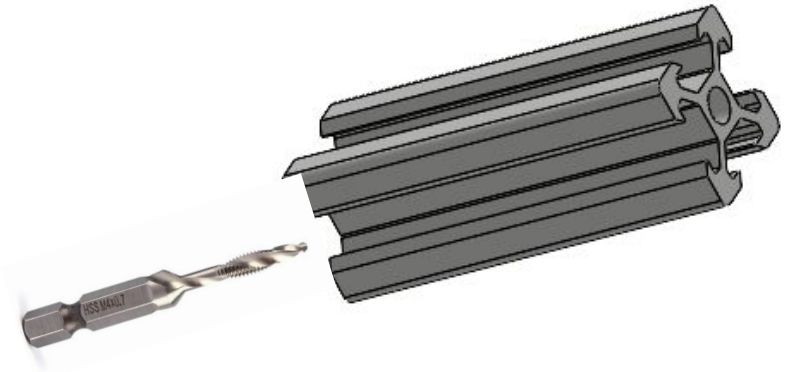
Attach the motor bracket assembly to the aluminum extrusion without fully tightening the screws



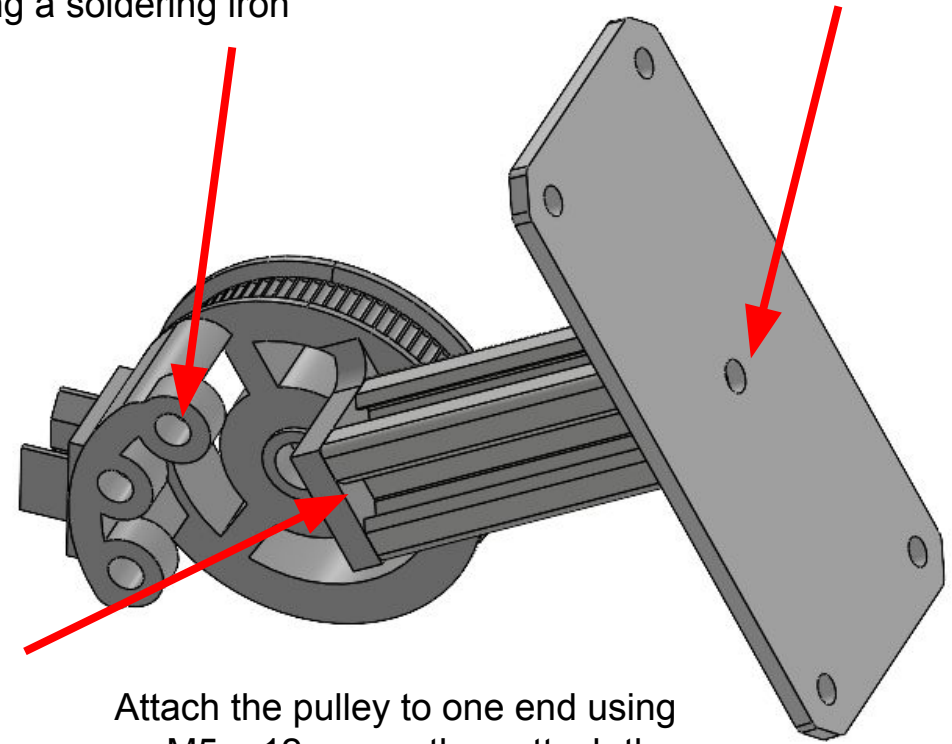


Secure the aluminum extrusion of the motor assembly to the head support using an M5 × 20 screw through the bearing spacer.

Cut a 60 mm length of 2020 aluminum extrusion and tap both ends with an M5 thread

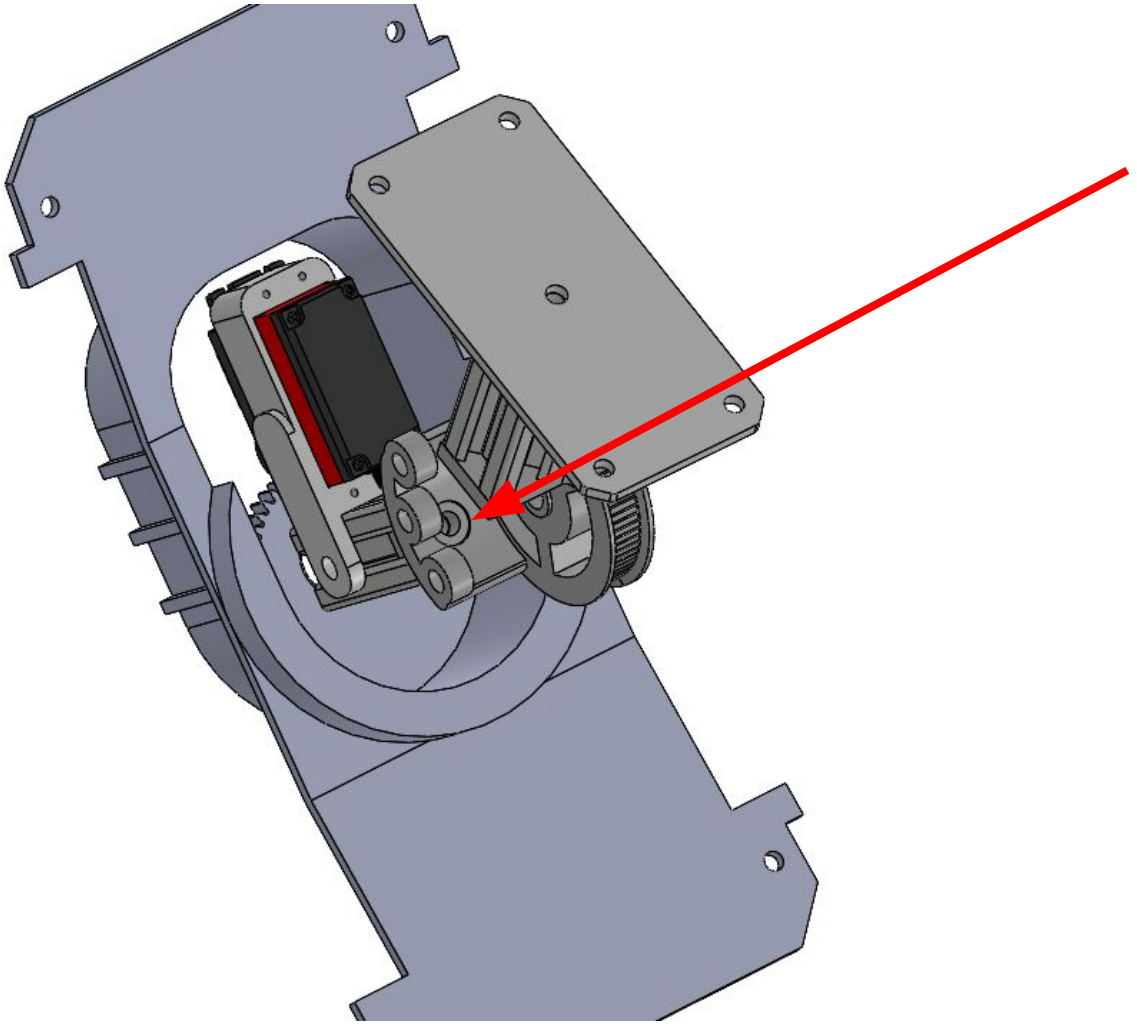


Insert three M4 threaded inserts using a soldering iron

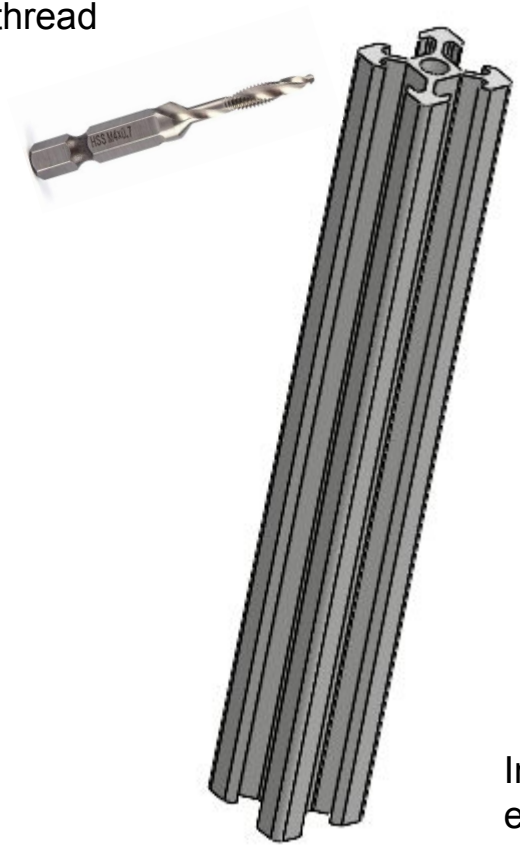


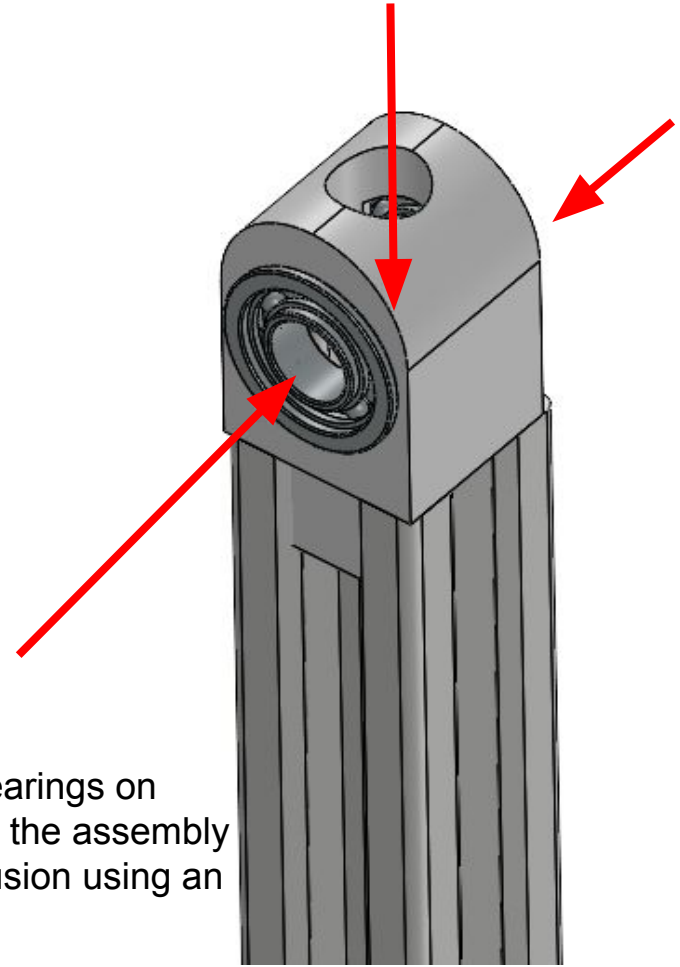
Attach the pulley to one end using an M5 × 12 screw, then attach the counterweight support to the other end using another M5 × 12 screw

Secure the head assembly to the pulley using an M5 × 12 screw



Cut a 150 mm length of 2020 aluminum extrusion and tap both ends with an M5 thread

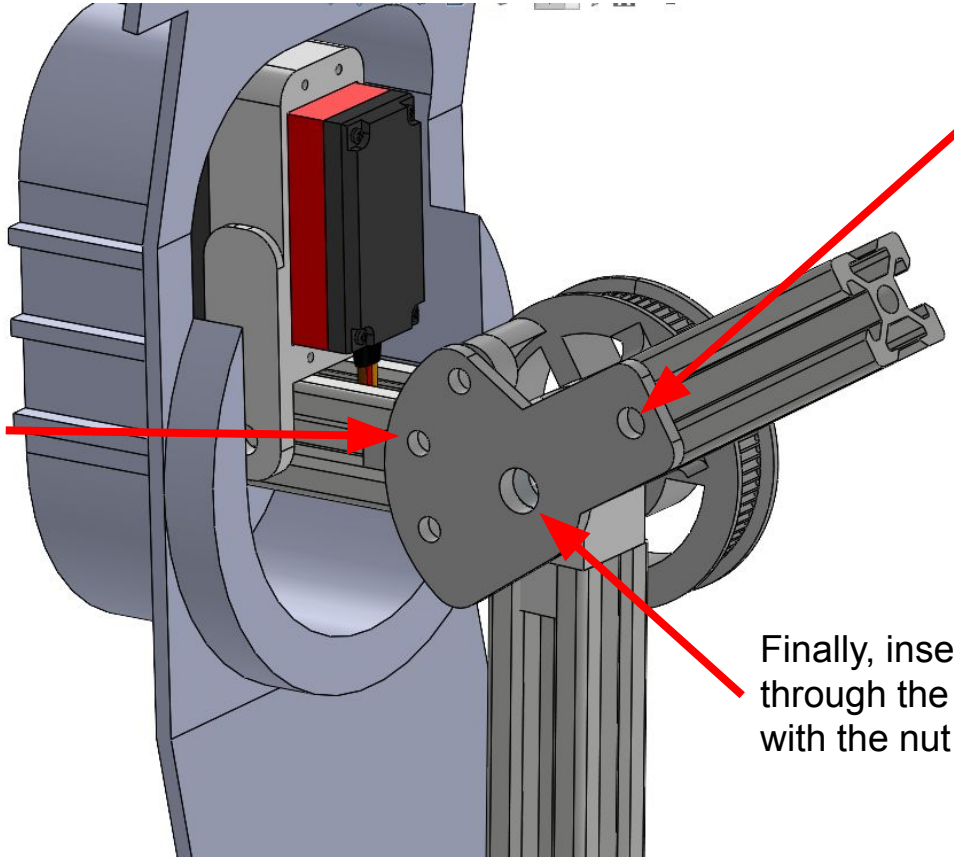




Install two F688ZZ bearings on each side and secure the assembly to the aluminum extrusion using an M5 × 12 screw.

Assemble the head section with the pivot section using the locking plate

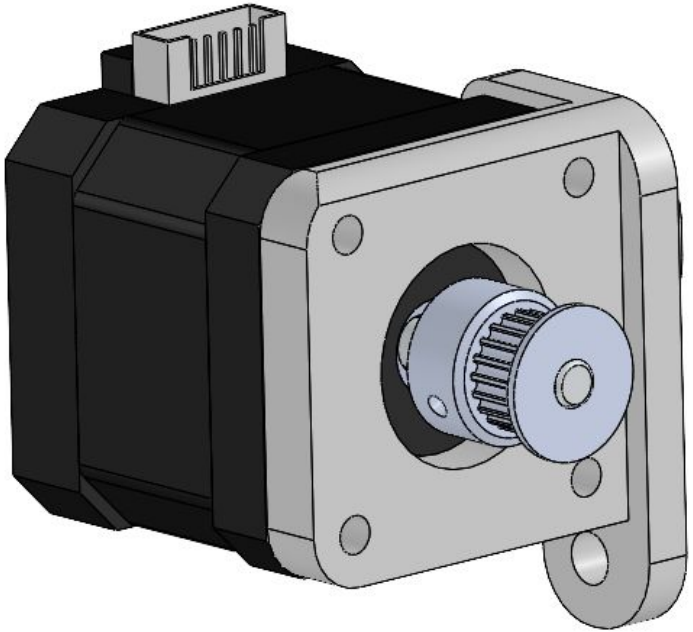
Install one of the extrusion screws supplied with the brackets to reinforce this section



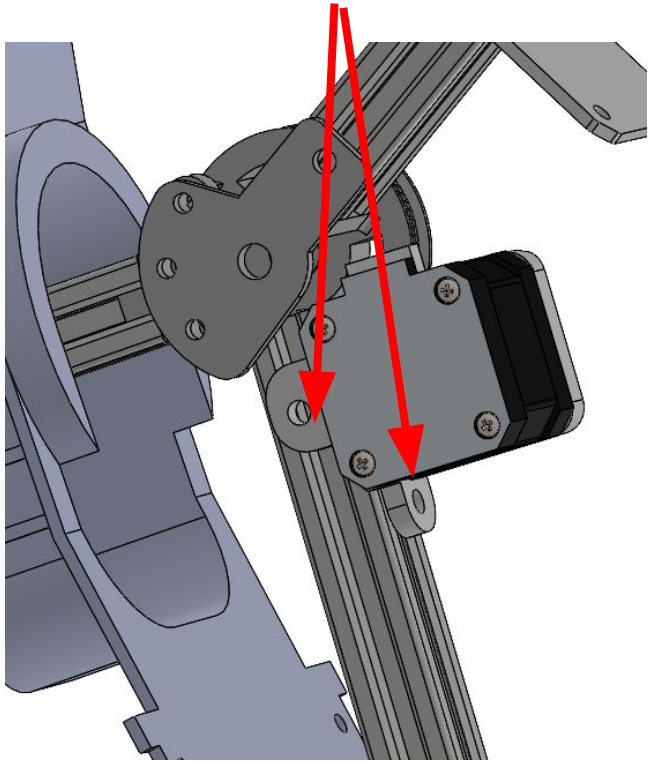
Secure the assembly using 3 M4 × 10 screws

Finally, insert the M8 × 40 screw through the pivot axis and secure it with the nut on the opposite side

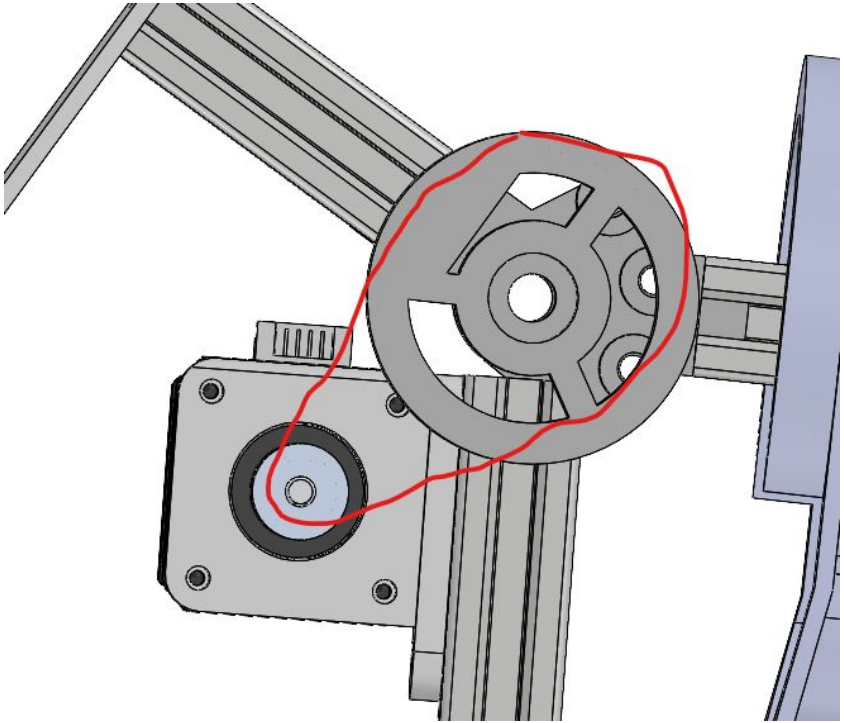
Secure the NEMA 17 motor to the NEMA motor bracket using four M3 × 16 screws, then attach the pulley to the motor shaft.



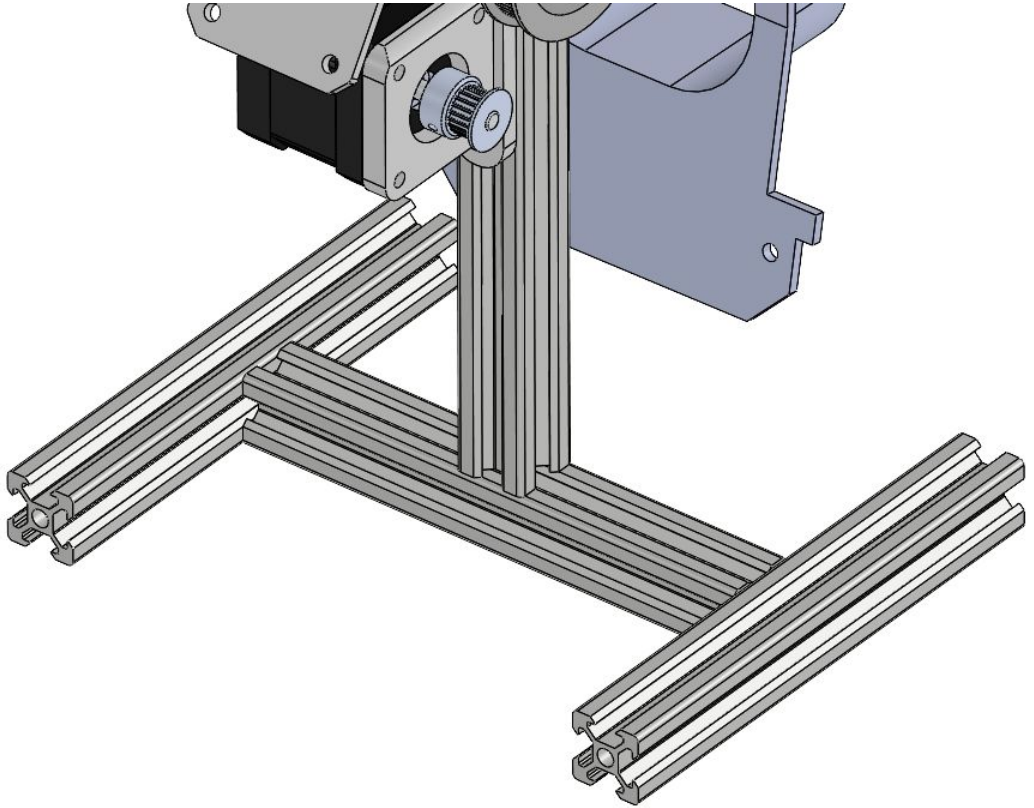
Secure the NEMA motor bracket to the 150 mm aluminum extrusion using two screws supplied with the extrusion brackets



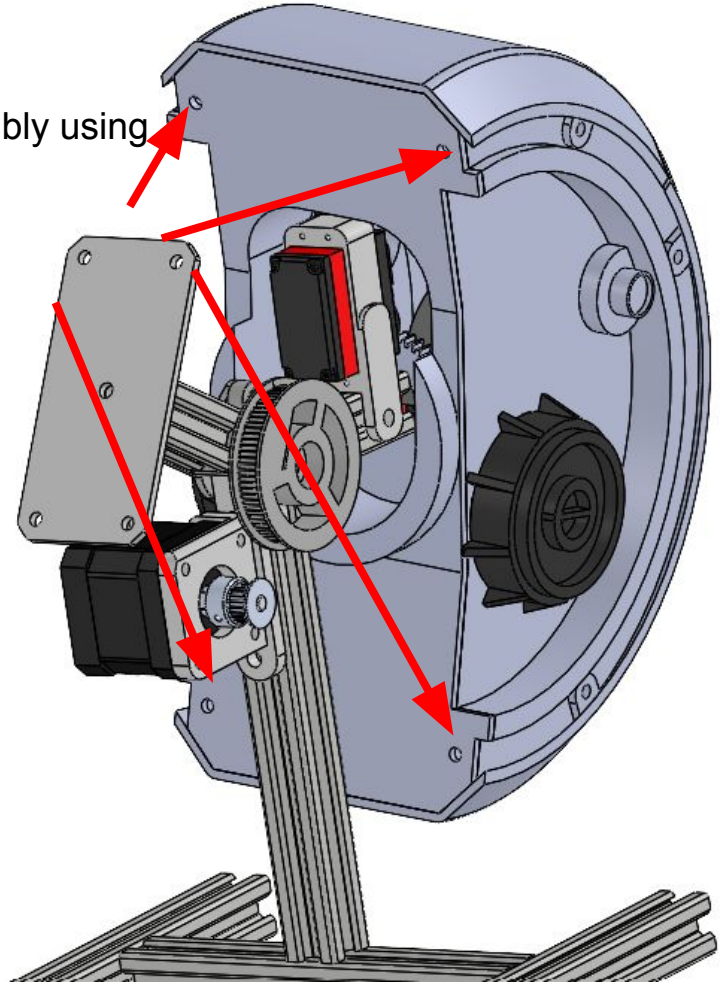
Install the GT2 belt to connect the two pulleys, then adjust the position of the motor bracket to set the correct belt tension



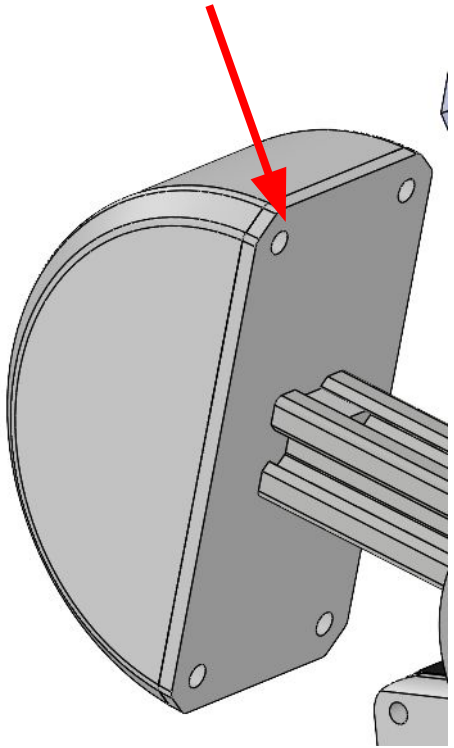
Assemble the base as shown in the image below using aluminum extrusions and brackets



“Install the head assembly using four M5 × 12 screws



Fill the counterweight with approximately 1 kg of material (nuts, sand, pellets, etc.), then close it using four M4 × 10 screws inserted into the previously installed M4 threaded inserts.



Your GLaDOS is now fully assembled!

